seq2seq_denoising.py — technical

`src/moe_pruning/tasks/seq2seq_denoising.py · 87 lines · moe_pruning.tasks.seq2seq_denoising`

Purpose

Encoder-decoder objective.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`class Seq2SeqDenoisingTask`

lines 20-87 · bases TaskObjective

Encoder-decoder objective with T5 span-corruption evaluation.

`Seq2SeqDenoisingTask.tokenize_calibration`

```
def tokenize_calibration(self, records, tokenizer, max_source_length, max_target_length)
```

lines 29-30

`human` is the input/prompt, `response` the teacher-forced target.

`Seq2SeqDenoisingTask.collate`

```
def collate(self, items, pad_token_id)
```

lines 32-33

Right-padded batch with per-stack real-token masks.

`Seq2SeqDenoisingTask.eval_settings`

```
def eval_settings(self, cfg) -> dict
```

lines 35-43 · settings EVAL_MEAN_SPAN_LENGTH, EVAL_NOISE_DENSITY, EVAL_SEED, MAX_SOURCE_LENGTH, MAX_TARGET_LENGTH

Task-specific settings that determine the evaluation examples.

`Seq2SeqDenoisingTask.build_eval_example`

```
def build_eval_example(self, example_id, text, tokenizer, cfg)
```

lines 45-50 · settings EVAL_MEAN_SPAN_LENGTH, EVAL_NOISE_DENSITY, EVAL_SEED, MAX_SOURCE_LENGTH, MAX_TARGET_LENGTH

One evaluation example (must expose `example_id`, `input_ids`, `labels`).

`Seq2SeqDenoisingTask.token_nll`

```
def token_nll(self, logits, labels)
```

lines 52-59

Per-example (sum of token NLL, number of scored tokens).

1. Move labels to the logits' device (last GPU when sharded); per-token cross-entropy with `ignore_index=-100`, `reduction='none'`; multiply by the label mask; sum per example; also return token counts.

`Seq2SeqDenoisingTask.predictions`

```
def predictions(self, logits, labels)
```

lines 61-65

Per-example argmax token at every scored position (same order as `token_nll`).

1. Argmax over vocabulary at positions whose label is not -100.

`Seq2SeqDenoisingTask.qualitative`

```
def qualitative(self, model, tokenizer, examples, device, cfg)
```

lines 68-87 · settings MAX_TARGET_LENGTH

`[{"example_id", "inputs": {label: text}, "prediction"}]` from greedy generation.

1. For each example: `generate` greedily from the corrupted input (max new tokens = `MAX_TARGET_LENGTH`); return original document, corrupted input, target and prediction as text.